

MORA, MN, USA

WMO: 727475

Lat: 45.886N

Lon: 93.272W

Elev: 309

StdP: 97.67

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 04940

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.7	-24.1	-32.4	0.2	-27.4	-28.7	0.3	-23.5	9.8	-10.5	8.7	-9.0	1.0	320	0.548

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.8	31.0	22.5	29.1	21.1	27.4	19.8	23.7	28.6	22.4	27.3	21.3	25.4	4.0	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
22.2	17.6	27.3	21.1	16.3	25.7	19.8	15.1	24.2	72.4	28.5	67.3	27.0	63.2	25.1	28.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.6	7.6	6.6	-31.2	33.3	3.7	1.9	-33.9	34.6	-36.1	35.7	-38.2	36.8	-40.9	38.2		
(5)			-31.1	25.6	3.7	1.3	-33.7	26.6	-35.9	27.3	-37.9	28.1	-40.6	29.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	5.8	-10.8	-8.6	-1.7	5.7	12.3	17.7	20.4	18.9	14.6	7.5	0.5	-7.3	(6)
(7)		DBStd	12.02	7.41	7.15	6.61	5.10	4.48	3.75	3.16	3.16	4.60	4.62	5.24	6.57	(7)
(8)		HDD10.0	2646	646	520	366	145	26	1	0	0	11	107	288	537	(8)
(9)		HDD18.3	4748	904	753	622	378	194	54	14	31	128	337	536	796	(9)
(10)		CDD10.0	1127	0	0	2	17	97	232	322	277	150	28	2	0	(10)
(11)		CDD18.3	189	0	0	0	1	7	36	78	49	17	1	0	0	(11)
(12)		CDH23.3	1882	0	0	0	9	121	388	747	441	166	11	0	0	(12)
(13)		CDH26.7	508	0	0	0	2	32	107	220	108	39	1	0	0	(13)
(14)	Wind	WSAvg	2.4	2.5	2.6	2.6	2.8	2.7	2.2	2.0	1.8	2.2	2.6	2.6	2.5	(14)
(15)	Precipitation	PrecAvg	721	22	18	38	61	88	104	96	79	62	42	23		(15)
(16)		PrecMax	1052	74	56	110	144	195	227	229	341	195	216	157	76	(16)
(17)		PrecMin	371	2	0	6	2	22	23	23	18	11	6	1	3	(17)
(18)		PrecStd	166	17	14	23	33	40	49	49	52	47	50	33	17	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.0	9.1	18.0	24.0	30.1	32.6	32.9	32.2	30.2	24.9	17.7	7.6	(19)
(20)			MCWB	3.5	5.9	11.4	13.9	18.9	23.3	24.3	22.6	20.9	16.4	11.9	4.1	(20)
(21)		2%	DB	2.6	6.1	13.0	21.1	26.4	29.9	31.1	29.9	27.4	21.3	13.7	4.7	(21)
(22)			MCWB	0.7	3.5	8.0	11.8	16.0	21.2	23.2	21.7	19.9	15.0	9.8	2.5	(22)
(23)		5%	DB	1.1	2.9	10.2	17.7	23.8	27.4	29.7	27.4	25.1	17.8	11.1	2.4	(23)
(24)			MCWB	0.0	1.1	6.9	9.9	15.1	19.1	21.9	20.1	18.2	12.8	7.5	1.1	(24)
(25)		10%	DB	-0.1	1.4	7.3	15.1	21.3	25.5	27.3	26.0	22.8	15.2	7.9	1.2	(25)
(26)			MCWB	-0.9	-0.1	4.0	8.3	13.8	18.1	20.5	19.3	17.2	10.7	5.2	0.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.1	6.1	13.9	15.3	21.0	24.6	25.9	24.4	23.1	18.9	13.2	5.3	(27)
(28)			MCDWB	6.4	9.3	17.4	22.0	27.2	30.5	31.2	29.5	27.7	22.6	16.1	6.8	(28)
(29)		2%	WB	1.0	3.7	9.8	12.4	18.0	22.5	24.2	22.8	21.3	15.9	10.4	2.7	(29)
(30)			MCDWB	2.1	5.9	13.0	18.5	23.5	27.9	28.9	27.5	25.3	19.8	13.4	4.3	(30)
(31)		5%	WB	-0.1	1.5	6.8	10.6	16.5	20.8	22.8	21.6	19.8	13.5	7.6	1.2	(31)
(32)			MCDWB	0.6	3.0	10.4	16.8	21.6	25.4	27.7	25.7	23.8	17.6	10.2	2.2	(32)
(33)		10%	WB	-1.3	0.0	4.2	8.9	14.9	19.4	21.5	20.4	18.4	11.6	5.6	0.1	(33)
(34)			MCDWB	-0.4	1.0	7.1	14.6	20.2	23.8	25.9	24.1	22.2	14.5	8.4	0.8	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	10.1	10.8	10.5	11.4	12.6	11.9	11.8	11.7	11.8	10.2	8.5	8.4	(35)
(36)			MCDBR	9.9	10.9	13.6	16.8	16.8	14.3	13.7	13.7	14.3	15.0	13.5	9.8	(36)
(37)			MCWBR	8.5	8.6	9.0	9.5	8.6	7.1	6.9	7.1	7.9	9.2	9.1	8.0	(37)
(38)		5% WB	MCDBR	9.0	9.8	13.2	15.3	14.0	12.2	12.4	11.8	11.9	12.8	12.1	8.5	(38)
(39)			MCWBR	8.0	7.8	9.1	9.3	8.1	6.9	7.0	6.9	7.5	8.6	8.8	7.2	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.257	0.266	0.312	0.370	0.401	0.387	0.383	0.395	0.366	0.314	0.286	0.261	(40)	
(41)		taud	2.392	2.364	2.364	2.301	2.263	2.339	2.374	2.352	2.405	2.546	2.508	2.419	(41)	
(42)		Ebn at Noon	863	923	919	887	867	879	878	853	849	860	823	817	(42)	
(43)		Edh at Noon	70	89	104	122	131	122	117	115	100	75	64	62	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.66	2.76	3.86	4.64	5.33	5.87	6.22	5.19	3.98	2.47	1.55	1.16	(44)	
(45)		RadStd	0.18	0.25	0.35	0.44	0.44	0.42	0.42	0.22	0.39	0.28	0.33	0.14	0.14	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	-0.82	N/A	N/A	-156	-76
(47)	Regional (58 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page